

Eight great ideas in Computer Architecture

People building systems need to account for the possibilities / changes from start to finish, in order to not create out of date systems.
Years

(1) Moore's law

- Integrated circuit resources double every 16-24 months

(2) Use of abstraction to simplify design

- Captures a design at different levels of complexity for simpler, less cluttered representation of a problem
 - + Can be used to enable modularity
 - Components tested and developed in solidarity, only interacts externally through its own "interface"
 - + Enables portability
 - Can be used on a variety of computer architectures without significant alteration / modification
 - + Allows for improving efficiency through abstraction of low level details
 - + Allows for scalability
 - Independence allows for easy migration of programs to more powerful platforms to account for more requirement for resources
 - + Leads to industry standardization
 - Allows for compability between many components / devices
 - + Security
 - Helps improve security by isolating elements from a system
 - + Learning and education
 - No necessary requirement for understanding low level, just need to know common concepts.

(3) Amdahl's law

+ Make the common case fast

- Tends to enhance performance better than optimisation of the rare case
- Common case is often less complex than the rare case
- Speed-up in parallel systems

(4) Performance via Parallelism

+ Design for running tasks in parallel

⊖ Concurrency

- Ability to run/manage multiple tasks being executed at overlapping time intervals
- Parallelism

(5) Performance via Pipelining

+ Fire "bucket brigade" idea

+ Assembly line - every process one after another

+ Speed determined by slowest component

(6) Performance via Prediction

+ Sometimes asking for forgiveness > asking for permission

+ Faster to guess and start work while waiting for answer assuming not too expensive to recover from an incorrect guess and prediction is relatively accurate

↑
Cache miss?

(7) Dependability by redundancy

+ Redundant components to

- take over if component fails

- detect failures

+ Backup for failures

+ Quorum?

(8) Hierarchy of Memories

- Higher Capacity
 - Faster
 - Cheaper
- } Choose 2

Available memory affects size of possible problems able to solve
Cost of memory ^{mainly} affects cost of computer